

Last time:

~ Ito Processes: $dX_t = \mu_t dt + \sigma_t dW_t$

~ Ito's Lemma:

$$df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} (dX_t)^2$$

where $(dX_t)^2$ computed using

$$(dW_t)^2 = dt, \quad dW_t dt = dt^2 = dW_t^3 = 0$$

Integration By parts:

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + dX_t dY_t$$

$$\text{-- SDEs: } dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t \quad (21)$$

§5.2: Existence & Uniqueness of solutions to SDEs.

e.g. $\sigma \equiv 0$, f^{nice} , linear

not trivial:

$$\frac{dX_t}{dt} = X_t^2, \quad X_0 = 1, \quad X_t = \frac{1}{1-t}$$

$\nearrow X_t \nearrow \infty$
 $\nearrow t \nearrow 1$

$$\frac{dX_t}{dt} = 3X_t^{2/3}, \quad X_t = 0$$

$$X_t = \begin{cases} 0 & t < 2 \\ (t-2)^3 & t \geq 2 \end{cases}$$

$\leadsto$ many solutions.

$\leadsto$ Lipschitz condition on μ .

Thm 5.5:

Let $T > 0$. Suppose $\mu, \sigma: [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$,

for s.t:

$$(i) |\mu(t, x)| + |\sigma(t, x)| \leq C(1 + |x|) \\ \forall t \in [0, T], x \in \mathbb{R}$$

$$(ii) |\mu(t, x) - \mu(t, y)| + |\sigma(t, x) - \sigma(t, y)| \\ \leq D|x - y| \quad \forall t \in [0, T], x, y \in \mathbb{R}$$

In addition, let Z be an r.v. s.t

$$E[Z^2] < \infty, Z \text{ indep of } (W_t)_{t \geq 0}$$

Then $\exists!$ t-continuous solution to

$$(21), X_0 = Z, \text{ s.t. } X_t \text{ depends only on } Z, (W_s) \text{ and } E\left[\int_0^T X_s^2 ds\right] < \infty$$

NB. High-dimensional version.

Sketch Proof:

Uniqueness: Suppose X_t, X'_t solve (21).

Write $\mu_s = \mu(s, X_s), \mu'_s = \mu(s, X'_s)$

$\sigma_s = \sigma(s, X_s), \sigma'_s = \sigma(s, X'_s)$

$$E[|X_t - X'_t|^2] = E\left[\left(\int_0^t (\mu_s - \mu'_s) ds + \int_0^t (\sigma_s - \sigma'_s) dW_s\right)^2\right] \\ = 2E\left[\int_0^t (\mu_s - \mu'_s) ds\right] + 2E\left[\int_0^t (\sigma_s - \sigma'_s) dW_s\right] \\ = 2E\left[\int_0^t (\mu_s - \mu'_s)^2 ds\right] + 2E\left[\int_0^t (\sigma_s - \sigma'_s)^2 ds\right] \\ = 2D(1+t) E\left[\int_0^t (X_s - X'_s)^2 ds\right] \\ = 2D(1+t) \int_0^t E[(X_r - X'_r)^2] dr$$

Write $m(t) = E[(X_t - X'_t)^2]$,

$$m(t) \leq (1+t)D^2 \int_0^t m(s) ds$$

Gronwall's Lemma: $\Rightarrow m(t) = 0$

Existence: $Y_r^{(0)} = X_0$

$$\text{define } Y_r^{(k+1)} = X_0 + \int_0^r \mu(s, Y_s^{(k)}) ds \\ + \int_0^r \sigma(s, Y_s^{(k)}) dW_s$$

Then as above, get

$$E[(Y_t^{(k)} - Y_t^{(k+1)})^2] \leq 2(1+t)D^2 \times \\ \int_0^t E[(Y_s^{(k)} - Y_s^{(k-1)})^2] ds \\ \text{for } k \geq 1$$

$$\begin{aligned}
& \text{and } \mathbb{E}[(Y_t^{(1)} - Y_t^{(0)})^2] \\
& \leq 2C^2 t^2 (\mathbb{E}[X_0^2] + 1) \\
& \quad + 2C^2 t (\mathbb{E}[X_0^2] + 1) \\
& \leq A_1 t
\end{aligned}$$

Induction:

$$\begin{aligned}
& \mathbb{E}[(Y_t^{(k)} - Y_t^{(k+1)})^2] \\
& \leq \frac{A_2 t^{k+1}}{(k+1)!}
\end{aligned}$$

Summing

$$\begin{aligned}
& \mathbb{E} \left[\int_0^T (Y_t^{(n)} - Y_t^{(m)})^2 dt \right]^{1/2} \\
& \leq \dots \leq \sum_{k=m}^n \left(\int_0^T \frac{A_2 t^{k+1}}{(k+1)!} dt \right)^{1/2}
\end{aligned}$$

$\rightarrow 0$ as $m, n \rightarrow \infty$

$\Rightarrow Y^{(n)}$ a Cauchy sequence,
 $\Rightarrow$ limit exists.
 $\square$

§6. Diffusions

A solution to:

$$dX_t = \underbrace{\sigma(t, X_t)}_{\text{diffusion coeff.}} dW_t + \underbrace{\mu(t, X_t)}_{\text{drift coeff.}} dt$$

is a (time-inhomogeneous) diffusion

Defⁿ A (time-homogeneous) diffusion is a solution to:

$$dX_t = \sigma(X_t) dW_t + \mu(X_t) dt, \quad (26)$$

$t \geq s, X_s = x$

where $|\mu(x) - \mu(y)| + |\sigma(x) - \sigma(y)| \leq D|x - y|$

Denote $(X_t^{s,x})_{t \geq s}$, unique (strong) solution.
 $X^{0,x} = X^x$

$$\text{Expect } (X_{s+t}^{s,x})_{t \geq 0} = (X_t^x)_{t \geq 0}$$

(as a prob. meas.)

Write $\mathbb{E}^x[\cdot]$ for expectation using $X^{0,x}$

Thm [Markov Property]

$$\mathbb{E}[f(X_{t+h}) | X_t] = \mathbb{E}^{X_t}[f(X_h)]$$

Idea:

$$\begin{aligned} X_t &= X_0 + \int_0^t \sigma(X_u) dW_u + \int_0^t \mu(X_u) du \\ &= \left(X_0 + \int_0^s \sigma(X_u) dW_u + \int_0^s \mu(X_u) du \right) \\ &\quad + \int_s^t \sigma(X_u) dW_u + \int_s^t \mu(X_u) du \end{aligned}$$

X_s (indicated by a red arrow pointing to the first term in the second line)

Defⁿ A random time $\tau: \Omega \rightarrow [0, \infty]$
 is a stopping time if
 $\{\tau \leq t\} \in \mathcal{F}_t$ i.e.
 know at time t if τ has
 happened.

Thm [Strong Markov Property]

If X_t is an Itô diffusion,
 $\tau < \infty$ a.s.) a stopping time

$$\mathbb{E}^x[f(X_{\tau+h}) | \mathcal{F}_\tau] = \mathbb{E}^{X_\tau}[f(X_h)]$$

§6.1 Generators & The Dynkin Formula

Defⁿ 6.5 Let X_t be an Itô diffusion
 on $\mathbb{R}^n$. The (infinitesimal)
generator A of X is given by

$$Af(x) = \lim_{h \downarrow 0} \frac{\mathbb{E}^x[f(X_h)] - f(x)}{h} \quad x \in \mathbb{R}^n.$$

The set of functions where Af exists
 at all $x \in \mathbb{R}^n$ is D_A .

Remark 6-6

Remark 2.6
If U_n a sequence of open sets, $U_n \supset U_{n+1}$, $\{x\} = \bigcap_n U_n$

then:

$$Af(x) = \lim_{n \rightarrow \infty} \frac{E[f(X_{\tau_n}) - f(x)]}{E[\tau_n]}$$

$$\tau_n = \inf \{ t \geq 0 \mid X_t \notin U_n \}$$

Thm: Suppose X solves (26). If

$$f \in \underline{C}_0^2(\mathbb{R}), f \in \mathcal{D}, \text{ and}$$

spat support $\Delta f(x) = \varphi(x) \frac{\partial f}{\partial x} + \frac{1}{2} \cdot \sigma(x)^2 \frac{\partial^2 f}{\partial x^2}$

Proof $[n=d=1]$

$$\begin{aligned} \text{Itô: } f(X_t) &= f(x) + \int_0^t \frac{\partial f}{\partial x}(X_s) \mu(X_s) ds \\ &\quad + \frac{1}{2} \int_0^t \frac{\partial^2 f}{\partial x^2}(X_s) \sigma(X_s)^2 ds \\ &\quad + \int_0^t \frac{\partial f}{\partial x}(X_s) \sigma(X_s) dW_s \end{aligned}$$

Taking expectations:

Taking expectation from:

$$\mathbb{E}[f(X_t) - f(x)] = \mathbb{E} \left[\int_0^t \left(\frac{\partial f}{\partial x}(X_s) + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(X_s) \right) ds \right] + \mathbb{E} \left[\int_0^t \frac{\partial^2 f}{\partial x^2}(X_s) ds \right]$$

divide by t , take

$$= 0$$

Divide by t , take
limits as $t \rightarrow \infty$

$$Af(x) = \lim_{t \rightarrow 0} \frac{E^x[f(X_t)] - f(x)}{t} = p(x) \frac{\partial f}{\partial x} + \frac{1}{2} \sigma(x)^2 \frac{\partial^2 f}{\partial x^2}$$

Thm : [Dirichlet's Lemma]

$$f \in C_0^2, \quad \tau \text{ a stopping time.}$$

$$E^x[f(X_T)] = f(x) + E \left[\int_0^T A f(X_s) ds \right]$$

Example, d -dimensional β .M. $\leftarrow$ Reflection

$$A f(x) = \frac{1}{2} \sum_{i=1}^d \frac{\partial^2 f}{\partial x_i^2}$$



$$\tau_D = \inf \{t \geq 0 \mid X_t \in D\}$$

$$w(x) := \mathbb{E}^x [g(X_{t_0})]$$

$$\text{if } u \in \mathcal{D}, \quad = \mathbb{E}^x \left[\mathbb{E}^{X_{\tau_u}} [g(X_{\tau_u})] \right]$$

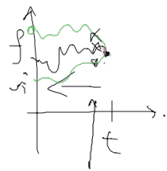
$$= E^x \left[w(X_{\tau_n}) \right]$$

$$\Rightarrow \underline{Aw(x)} = \lim_{u \rightarrow x} \frac{E^x[w(X_{tu}) - w(x)]}{E[-tu]}$$

§6.2 Kolmogorov's Backward Equation, Boundary Value Problems.

Thm [K.B.E.] X_t an Ito diffusion
on $\mathbb{R}^n$, generator A , $f \in C_0^2(\mathbb{R}^n)$

Set $u(t, x) := \mathbb{E}^x[f(X_t)]$



(i) $u(t, \cdot) \in \mathcal{D}_A$

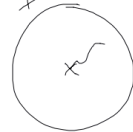
$$\begin{cases} \frac{\partial u}{\partial t} = Au & t > 0, x \in \mathbb{R}^n \\ u(0, x) = f(x) & x \in \mathbb{R}^n \end{cases} \quad (2.9)$$

(ii) If $w \in C^{1,2}$, solves (2.9),
then $w = u$.

Thm [Dirichlet-Poisson]

D bdd, open $\mathbb{E}^x[\tau_D] < \infty$

$\forall x \in \mathbb{R}^n$ $g \in C(\partial D)$, $h \in C(D)$



$$w(x) := \mathbb{E}^x \left[g(X_{\tau_D}) + \int_0^{\tau_D} h(X_s) ds \right]$$

If $w \in C^2(D) \cap C(\bar{D})$, then it is
unique solⁿ to

$$\begin{cases} Aw(x) = -h & x \in D \\ w(x) = g & x \in \partial D \end{cases}$$